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schism-toolbox: A MATLAB toolbox designed for the SCHISM ocean model

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Software

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Summary

Numerical modeling has become an indispensable tool for simulating hydrodynamic processes and supporting ecosystem research. However, configuring numerical models often involves a steep learning curve and requires substantial time investment in data pre- and post-processing. The Semi-implicit Cross-scale Hydroscience Integrated System Model (SCHISM), as one of the state-of-the-art ocean modeling systems (Y. J. Zhang et al., 2016), has been successfully applied to a wide range of oceanic and coastal environments (Amores et al., 2022; Mentaschi et al., 2023; Ye et al., 2020). This paper presents a newly developed MATLAB package, `schism-toolbox`, designed to lower the entry barrier for SCHISM users. The toolbox offers a flexible and efficient workflow for handling SCHISM input and output files, leveraging MATLAB's strong capabilities in vectorized operation and interactive data analysis.

Statement of need

SCHISM is a three-dimensional, hydrostatic, and baroclinic ocean modeling system grounded on unstructured grids (Y. J. Zhang et al., 2016). It is an advanced derivative of the original Semi-implicit Eulerian-Lagrangian Finite-Element (SELFE) model (Y. Zhang & Baptista, 2008) and distributed with an open-source Apache v2 license. This model has undergone significant enhancements, including extensions to large-scale eddy-resolving regimes and a seamless cross-scale capability spanning from small creeks to the open ocean.

Nowadays, SCHISM has been widely applied across a broad range of oceanographic settings worldwide, demonstrating robust skills in simulating hydrodynamic processes and supporting biogeochemical studies (Cai et al., 2023; Liu et al., 2018; Ye et al., 2020). The model has fostered a large and expanding user community, drawing researchers from diverse fields such as physical oceanography, ocean engineering, and marine ecology. Several tools have been developed to support SCHISM's pre- and post-processing workflows, including Python-based packages like `pyschism` and `pylibs`, as well as official Fortran utilities. However, MATLAB-based tools remain relatively scarce.

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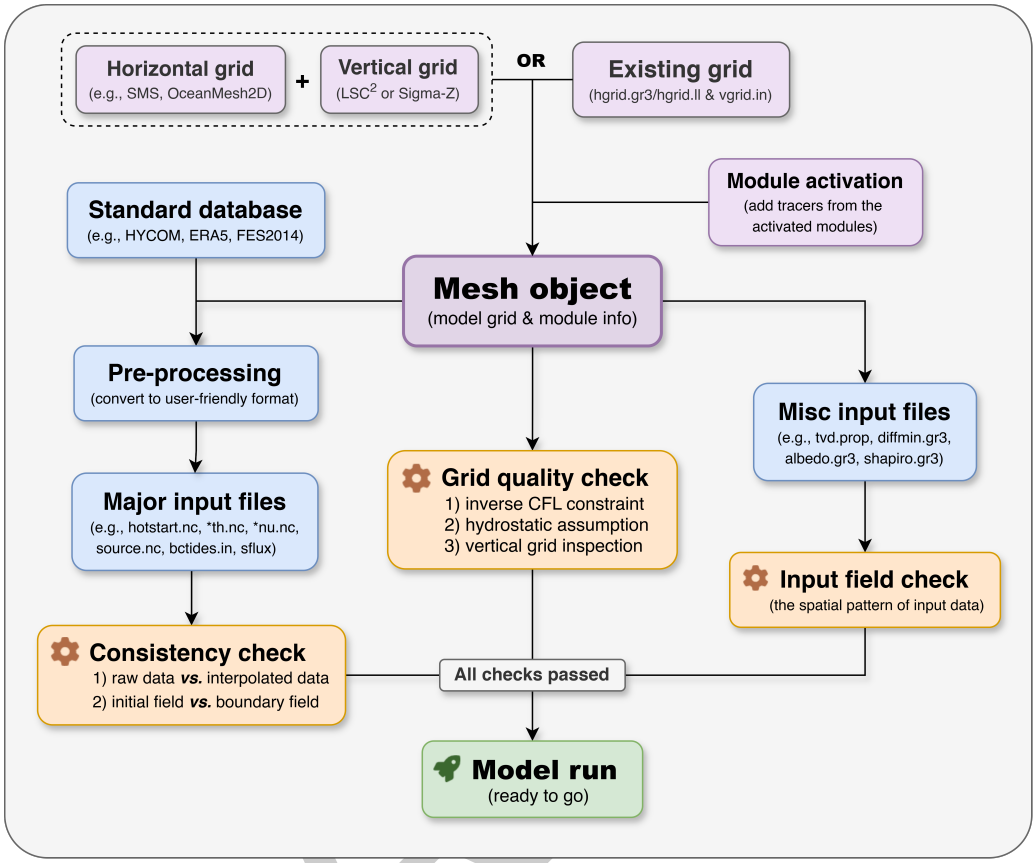

Figure 1: Typical workflow to prepare SCHISM input files based on the toolbox

To this end, `schism-toolbox` was developed to support the needs of the user community. Leveraging MATLAB's powerful vectorized operations and interactive visualization, it offers an efficient, user-friendly workflow for data processing and analysis, with minimal dependence on external packages. This workflow allows users to rapidly set up hydrodynamic or coupled hydrodynamic-biogeochemical simulations with SCHISM, provided that a standardized database is available.

Figure 1 illustrates a typical workflow of the toolbox, which adopts a modular design to improve usability and maintainability. During the setup process, grid information and activated modules are stored in a mesh object named `Mobj`, which globally manages the model configuration within the toolbox. Data preparation is generally organized into three modular steps: raw data extraction, format standardization (conversion to a user-friendly structure), and interpolation onto the SCHISM grid. A comprehensive quality check is implemented for both the model grid and input data to ensure the accuracy and consistency of the input files. In addition, the toolbox provides a suite of utility functions for working with unstructured grids, including variable visualization, gradient computation, contour line extraction, and more.

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